Q.No 8

Question description

Hassan enjoys jumping from one building to the next. However, he merely jumps to the next higher building and stops when there are none accessible. The amount of stamina necessary for a voyage is equal to the xor of all the heights Hassan leaps till he comes to a halt.

If heights are [1 2 4], and he starts from 1, goes to 2 stamina required is $1 \oplus 2 = 3$, then from 2 to 3. Stamina for the entire journey is $1 \oplus 2 \oplus 4 = 7$. Find the maximum stamina required if can start his journey from any building.

Constraints

$1 \le N \le 10^5$

$1 \le Height \le 10^9$

Input

First line: N, no of buildings.

Second line: N integers, defining heights of buildings.

Output

Single Integer is the maximum stamina required for any journey.

Explanation:

8

1 2 3 8 6 4 7 9

Considering the input given by you, 1st starting point is 1. From this building, Hassan will try to find the next higher building that is building 2 here and so on. Likewise he will complete all pass.

Pass 1: 1-> 2->3->8-> 9, XOR = 1

Pass 2: 2->3->8-> 9, XOR = 0

Pass 3: 3->8-> 9, XOR = 2

Pass 4: 8-> 9, XOR = 1

Pass 5: 6->7->9 , XOR = 8

Pass 6: 4->7->9 , XOR = 10

Pass 7: 7-> 9 XOR = 14

Pass 8: 9 XOR = 9

So, the highest XOR value is 14

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
1 2 3 8 6
```

EXPECTED OUTPUT

```
11
```

Test Case 2

INPUT (STDIN)

```
8
1 2 3 8 6 4 7 9
```

EXPECTED OUTPUT

```
14
```

Code:

```
#include <stdio.h>

int main()
{
int n;
int arr[1000000];
int st[1000000];
int stack[1000000];
```

```c
int top = -1;
int i, j;
int max = 0;

scanf("%d", &n);

for(i = 0; i < n; i++)
{
scanf("%d", &arr[i]);
}
for(i = n - 1; i >= 0; i--)
{
while(top != -1 && arr[stack[top]] <= arr[i])
{
top--;
}

if(top == -1)
{
st[i] = arr[i];
}
else
{
j = stack[top];

if(arr[i] < arr[j])
{
st[i] = arr[i] ^ st[j];
}
else
{
st[i] = arr[i];
}
}

stack[++top] = i;
}
for(i = 0; i < n; i++)
{
if(st[i] > max)
{
max = st[i];
}
}

printf("%d\n", max);

return 0;
}
```

Output:

Test _case 1;

```
5
1 2 3 8 6
11%
```

Test _case 2:

```
8
1 2 3 8 6 4 7 9
14%
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```